

Aaron Ebrey

Computer Games Development Student
Gameplay Programming Focus

Portfolio:

<https://clawedatom.github.io/AEPortfolio/>

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Personal Profile

Computer Games Development student at Manchester Metropolitan University with over 6 years of programming experience and 5 years working with game engines, primarily Unity and C#. Experienced in developing gameplay systems including player controllers, state machine based systems and procedural elements as a solo developer. Actively seeking a placement in the games development industry to apply technical skills within a professional studio environment.

Education

Manchester Metropolitan University (*Current*)

Bsc (Hons) Computer Games Development

A-Level - King Edwards VI College Stourbridge (2020-2022)

Awards: Computer Science (C), Chemistry (C), Physics (B), Core Maths (C)

GCSE - Perryfields High School (2015-2020)

Awards: 8 Total GCSEs

Development Experience and Projects

A range of development projects can be viewed via my portfolio, GitHub and YouTube channel, where I document my development process and technical progress over the past 5 years.

Across these projects, I have worked as a solo developer, designing and implementing core gameplay systems, including:

- Player controller and movement systems
- State machine driven systems
- World and level generation systems
- Gameplay mechanics and prototyping

Key Projects

Tunnel Train Game Prototype

2D prototype (Unity C#)

Developed a functional prototype during a one-week university game jam, focus on drilling tunnels from A to B using tile map destruction and systems driven gameplay.

Key Systems

- Designed and developed a tile map-based destruction system using Scriptable Objects to define tile types and behaviours.
- Implemented a drilling-based player controller with a stats system influencing gameplay.
- Applied BFS to ensure tunnels are valid
- Prioritized clean, readable code with an emphasis on technical proficiency and rapid iteration.

Playable Build: <https://zelldauron.itch.io/tunnel-digging-train-game> • **GitHub Source:**

<https://github.com/Clawedatom/Tunnel-Train-Game>

Alien Survivor

2D Wave Shooter – Team Project (Unity, C#)

Wave Shooter developed in a 7-person team, where I took the role of Lead Programmer. The project lasted 3 months, during which I developed core systems, built prototypes, and assisted other programmers throughout the development cycle.

Lead Programmer Responsibilities

- Assisting other programmers with complex development tasks.
- Offering design ideas and acting as a communication bridge between the design team and programmers.
- Integrating systems developed by other programmers.
- Fixing uses in areas such as enemy spawn weights, enemy AI and UI integration.
- Contributing to Trello and GitHub discussions as well as updates.

Key Systems

- **Wave-Based Enemies** — Multiple enemy types with unique AI behaviours.
- **Player Character & UI** — Movement, shooting, health/progression display, and general player systems.
- **Clean, Structured Code** — Ensured readability and maintainability to reduce confusion among team members.
- **Procedural Map Generation** — Grid-based system using Perlin noise to spawn obstacles, paired with a building system for turret placement.
- **Menu & Pause System** — Implemented a smooth game loop with transitions and UI responsiveness.

Key Contributions

- **Map Generation** – Grid-based tiles with Perlin noise to create rock clusters.
- **Building System** – System for placing turrets.
- **Turrets** – Turrets that target the closest enemy.
- **Systems Development** – Connected various systems throughout the sprint.

Greedy Little Goblin

Co-Hosted Game Jam Project – Fast-Paced Heist Game (Unity, C#)

Solo developed and published a fast-paced heist game during a student game jam that I co-hosted, balancing development responsibilities alongside organisational and community management tasks.

Key systems

- Designed and implemented a state driven character controller, including dynamic shrinking mechanics
- Developed gameplay systems that adjust collision, interaction and detection based on player size
- Implemented level structure and progression systems within strict time constraints
- Scoped and prioritised features to ensure a complete build by the deadline.
- Contributed to jam organisation, including setup and participant engagement

Playable Build: <https://zelldauron.itch.io/greedy-little-goblins> · GitHub Source: <https://github.com/Clawedatom/GreedyLittleGoblin>

Employment Experience

Warehouse Operative – Gadget, Oldbury

(Oct 2022 – Feb 2023)

- Worked effectively within a fast-paced team environment
- Demonstrated reliability, time management and adaptability under pressure

Technical Proficiencies

- Unity Engine & C# - 5 years
- Unreal Engine & C++ - 1+ years
- GitHub/Version Control – 3 years
- Microsoft Visual Studio – 5 years

Interests

Game Development – Regularly design and develop games with a focus on gameplay mechanics, systems design and technical problem solving.

Fitness – Consistently training in the gym for the past 5 years, demonstrating commitment and discipline.

References

Available on request.